Nishith Kotak

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GitHub Link: https://github.com/nishithkotak
Google Scholar Link: https://scholar.google.com/citations?user=mt6P4TgAAAAJ&hl=en
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Professional Summary

Skilled Researcher enthusiastic about supporting advancements in the fields of Data Analysis, Natural Language Processing and Recommendation Systems. Passionate about increasing knowledge to drive growth and needed improvements. Ready to apply knowledge and my experience in the research and development.

Skills

- Natural Language Processing
- Recommendation Systems
- Data Analysis
- Machine Learning
- Deep Learning
- ML Frameworks: TensorFlow, PyTorch
- Image Processing and Computer Vision
- Data Structures and Algorithms
- Programming Languages (Python, C++, R, SQL, C)
- EDA Tools: PowerBI, Tableau, Microsoft Excel

Experience

- **Pursuing Doctoral Programme** in the fields of NLP, Recommendation Systems and Optimization Techniques at Marwadi University, Rajkot, Gujarat
- **Assistant Professor** - Department of Information and Communication Technology, Marwadi University, Rajkot, Gujarat. [UGC-NET 2024 Qualified as Assistant Professor in Computer Science and Applications] [Sept, 2017-Present]

Academia

Examination	Institute Name	Board	Year of passing	Percentage(%)
Ph.D.	MU, Rajkot	MU	Ongoing	90.00%
MTech (ICT)	DA-IICT, Gandhinagar	DAIICT	2017	71.40%
BE (EC)	MEFGI, Rajkot	GTU	2015	82.30%
H.S.C	Vidyaniketan school, Rajkot	GSHSEB	2011	79.23%
S.S.C	St. Mary high school,Rajkot	GSEB	2009	81.23%

Major Project Works

1. **A Multi-Probe Evaluation Framework for Reviewer Matching in Scholarly Peer Review** [Ph.D. Thesis]

   [**Description:** The Multi-Probe Evaluation Framework (ReFORM) is designed to assess the consistency and reliability of reviewer-manuscript matchmaking systems in scholarly peer review. It introduces three complementary probes that measure alignment between manuscripts, reviewer profiles, and submitted reviews to detect potential mismatches or inconsistencies. By quantifying both direct expertise and contextual coherence, ReFORM offers a comprehensive, data-driven approach to evaluating the quality of automated reviewer assignment models.]
   Mentors: Dr. Bakul Gohel, Associate Professor, Marwadi University, India
   Dr. Arjav Bavarva, Associate Professor, Marwadi University, India

2. **Self Correcting Pen: A Pen that Suggests Correct Spellings** [MTech Thesis]
   [**Description:** The pen is designed such that it automatically gives suggestions of the words that the user is writing on the paper with smart pen resulting in error-free writings on paper.]
   Mentor: Dr. Anil K. Roy, Associate Professor, DA-IICT, India

3. **IoT Based: Smart Chair Indicating Comfortable Sitting Positions** [BE Final Year Project]
   [Description: The chair is made so smart that it will detect the sitting position of the user and notify him if his sitting posture is found uncomfortable. Also the chair is made user-defined.]
   Mentor: Prof. Ravi Bagree, Asst. Professor, MEFGI, India

Research Publications

1. M. Mithani, **N. A. Kotak,** T. Nahar, C. D. Parmar, "High-Performance Scientific Data Visualization Using VTK: Design and Evaluation of Rendering Pipelines", International Conference on Next Generation Communication & Information Processing, 2026 [Accepted, to be published]
2. ****N. A. Kotak,** B. Gohel, A. Bavarva**,** "C²Specter: Centroid-Constrained Supervised Contrastive Learning for Domain-Aware Reviewer Assignment, Scientometrics, Springer Nature [In-Communication]
3. ****N. A. Kotak**, B. Gohel, A. Bavarva, "Argument-Driven Keyphrase Extraction for Enhanced Scholarly Document Representation", Journal of Scientometric Research [In-Communication]
4. A. Singh, A. Raj, **N. A. Kotak,** C. D. Parmar, "MER-1M: A LARGE-SCALE HIGH-DIMENSIONAL BENCHMARK FOR MUSIC ENERGY AND AGENTIC AI PROFILING", International Journal of Arts and Technology [In-communication]
5. **N. A. Kotak,** A. Bavarva, C. D. Parmar, "Pedagogical Transformation through Integration of ICT in Higher Education: An Empirical", International Journal of Learning and Change [In-Communication]
6. Stanley Otieno, Nishith Kotak, Chandrasinh Parmar et al. Defect Detection and Classification on Hot-Rolled Steel Using Machine Learning and Deep Learning Algorithms: A Comparative Analysis, 23 August 2023, PREPRINT (Version 1) available at Research Square [https://doi.org/10.21203/rs.3.rs-3281494/v1]
7. Padia Darshan, Sharma Aryavardhan, Bharmal Mustafa, Chandarana Foram, **Nishith Kotak,** "How effective is game based learning for teaching graph theory concepts?: A Case Study of the Treasure Hunt Game", 11th IEEE Humanitarian Conference (HTC), 2023
8. Parmar, Chandrasinh, **Kotak, Nishith**, Sorathiya, Vishal and Patel, Shobhit K.. "A comprehensive study of plant pest and disease detection using different computer vision techniques". *Artificial Intelligence for Signal Processing and Wireless Communication*, edited by Abhinav Sharma, Arpit Jain, Ashwini Kumar Arya and Mangey Ram, Berlin, Boston: De Gruyter, 2022, pp. 47-68. https://doi.org/10.1515/9783110734652-003
9. **Kotak N.**, Roy A.K., Dasgupta S., Ghosal T., "*A Consistency Analysis of Different NLP Approaches for Reviewer-Manuscript Matchmaking*", 2021 In: Ke HR., Lee C.S., Sugiyama K. (eds) Towards Open and Trustworthy Digital Societies. ICADL 2021. Lecture Notes in Computer Science, vol 13133. Springer, Cham. https://doi.org/10.1007/978-3-030-91669-5_22
10. **N. Kotak**, A. K. Roy, S. Dasgupta and T. Ghosal, "*Consistency Analysis of NLP Approaches for a Conference Reviewer-Manuscript Match-Making System*", 2021 IEEE 18th India Council International Conference (INDICON), 2021, pp. 1-6, doi: 10.1109/INDICON52576.2021.9691571.
11. Batoul Aljaddouh, **Nishith A. Kotak**, "*Document Text Classification Using Support Vector Machine*", International Journal of Engineering Development and Research (IJEDR), ISSN:2321-9939, Volume.8, Issue 1, pp.138-142, January 2020
12. A. B. Jani, **N. A. Kotak** and A. K. Roy, "*Sensor Based Hand Gesture Recognition System for English Alphabets Used in Sign Language of Deaf-Mute People*", 2018 IEEE SENSORS, New Delhi, 2018, pp. 1-4. DOI: 10.1109/ICSENS.2018.858957.
13. **Nishith A. Kotak**, Anil K. Roy "*An accelerometer based handwriting recognition of English alphabets using basic strokes*", Proc. of the 2017 IEEE Region 10 Conference (TENCON), Malaysia, November 5-8, 2017 pp.111-116, ISSN: 2159-3450,DOI: 10.1109/TENCON.2017.8227846
14. ****Kotak N.**, Bavarva A., Parmar C D., "*Learner-Centred Approach Assessment Framework (LAF) for Indian Higher Education System*"
15. **Bhumi Trivedi, Chandrasinh Parmar, **N. A. Kotak**, "Role of Machine Learning in IR-Drop Prediction in the Circuit Layout" [To be Communicated]
16. ****N. A. Kotak**, B. Gohel, A. Bavarva**,** "Matching Manuscripts to Experts: A Review of Intelligent Peer-Review Assignment Systems" [To be Communicated]
    ** *To be Communicated/In-Communication*

Patents Published

1. **Patent Published** with title of invention "Method for Converting Handwritten Characters into Machine-Readable Instructions". Application Number: 202421033236
2. **Patent Published** with title of invention "Method for Evaluating Typing Accuracy with Adaptive Difficulty Levels". Application Number: 202421033381
3. **Patent Published** with title of invention "Vehicle Undercarriage Surveillance Apparatus". Application Number: 202421033237
4. **Patent Published** with title of invention "Adaptive Traffic Management System". Application Number: 202421070316
5. **Patent Published** with title of invention "Student Assistant Device for Scholastic Improvement". Application Number: 202521014339
6. **Patent Published** with title of invention "Access and Productivity Tracking Device for Regulating Usage of Laptop Gadgets". Application Number: 202521014338
7. **Patent Published** with title of invention "Essential Service Management System for Trains". Application Number: 202521014382
8. **Patent Published** with title of invention "Public Transportation Management System". Application Number: 202521015110
9. **Patent Published** with title of invention "Wildlife Based Safety System". Application Number: 202521015109
10. **Patent Published** with title of invention "User-Centric Focus Management System". Application Number: 202521024286
11. **Patent Published** with title of invention "Goods Management Device For Vehicle". Application Number: 202521031749
12. **Patent Published** with title of invention "Food Residues Straining Assistive Device". Application Number: 202521031758
13. **Patent Published** with title of invention "Multi-Functional Writing Instrument". Application Number: 202521040542
14. **Patent Published** with title of invention "Multi-Modal Typing Assistive System". Application Number: 202521052017
15. **Patent Published** with title of invention "Gesture Controlled Control Drone". Application Number: 202521050919
16. **Patent Published** with title of invention "Multimodal Language Translation And Communication Wearable Glasses". Application Number: 202521056016

Professional Achievements

1. **Qualified UGC-NET** as Assistant Professor in Computer Science and Applications in 2024 with 98.49 Percentile.
2. **Delivered an Expert Talk** on "Hands-On implementation of AI algorithms over Genomics Data" in International Workshop on Advancements of Genomics Using AI (IWAGAI-2025).
3. **Reviewer** in IEEE International conference on Artificial Intelligence and Smart Technologies for Sustainability (AISTS-2025)
4. **Reviewer** in IEEE Region 10 Symposium TENSYMP-2022 and TENSYMP-2023
5. **Reviewer** in IEEE Region 10 Symposium TENSYMP-2022 and TENSYMP-2023.
6. **Reviewer** in IEEE 11th Humanitarian Technology Conference HTC-2023.
7. **Reviewer** in IEEE India Council International Conference INDICON-2019.
8. **Reviewer** in International Conference on Transformations in Engineering Education ICTIEE-2020.
9. **Mentored** the project "Designing an integrated coding and event management platform", which got awarded **1st Rank**, in the Hackentine 2025 national hackathon by 10X club of Coding Ninjas.
10. **Mentored** the project "Driver Drowsiness and Safe Alarming Distance System", which got awarded **2nd Rank**, in the international Project Contest MYOSA conducted by IEEE SENSORS in Oct. 2018.
11. **Mentor** of a **funded project** under NewGen IEDC, programme by NSTEDB, DST, Government of India
12. **Certified** Google Data Analytics Professional, issued by Google with Coursera, endorsed by American Council of Education (ACE) and Foundation for International Business Administration Accreditation (FIBAA).
13. **Certified** IIEECP (IUCEE International Engineering Educators Certification Program) Engineering Educator.
14. **Member** of a research project group under **LeadingIndia.ai** in collaboration with Bennett University.
15. **Mentored** a team in Gujarat Smart India Hackathon, where the team was a finalist in 2018

16. **Mentored** a team in Gujarat Smart Industrial Hackathon where team was a finalist in 2019
17. **Faculty Coordinator** of 10X Club Chapter of Coding Ninjas at Marwadi University
18. **Faculty Coordinator** of Data Science Club, ICT, MU
19. **Faculty Coordinator** of Competitive Programming Club, ICT, MU
20. **Guided** 4 MTech students in Thesis and 18 BTech students in BTech Final Year Project.
21. **Guided** 150+ students in their academic course/subject related projects.
22. **Qualified** GATE-CS with GATE Score 449 in 2020
23. **Qualified** GATE-EC with GATE Score 500 in 2015
24. Overall (average) 89.00%+ feedback from 2000+ students from 2017-present.
25. Member of Syllabus Designing committee of ICT Engineering Department for Diploma, BTech and MTech

Teaching Experience

A. Major Subjects Taught at MU, Rajkot, Gujarat:
Served as *Course Instructor* in

1. a. Design and Analysis of Algorithms (Program Core, BTech 5th Sem, ICT) —4 times
   b. Algorithmic Complexity and Design (Program Core, MTech 1st Sem, ICT) — 3 times
2. a. Machine Learning Lab (Program Elective, BTech 6th Sem, ICT) — 5 times
   b. Machine Learning (Program Core, MTech 2nd Sem, ICT) — 3 times
3. Artificial Intelligence (Program Core, BTech 6th Sem, ICT) — 5 times
4. Optimization Techniques (Program Core, BTech 6th Sem, ICT) — 3 times
5. Introduction to R and R Studio (Program Core, BTech 2nd Sem, ICT) — 3 times
6. a. Neural Network and Deep Learning (Program Elective, MTech 3rd Sem, ICT) — 2 times
   b. Deep Learning (Program Elective, BTech 6th Sem, ICT) —3 times
7. Information Retrieval and Natural Language Processing (Program Elective, BTech 7th Sem, ICT) —3 times
8. Data Structures and Algorithms (Program Core, BTech 2nd Sem, ICT) — 3 times

B. Subjects Taught at DAIICT, Gandhinagar, Gujarat

1. *Course Instructor* in Introduction to Python Programming (Program Core, MSc.DS 1st Sem) —2 times
2. *Teaching Assistant* in Introduction to ICT Workshop (Program Core, BTech 1st Sem) — 2 times
3. *Teaching Assistant* in Engineering Materials (Program Core, BTech 1st Sem) —2 times
4. *Teaching Assistant* in Object Oriented Programming (Program Core, BTech 2nd (Summer) Sem) —1 time
5. *Teaching Assistant* in Systems Programming (Program Core, MSc.IT 2nd Sem) —1 time

Workshop/Certification Undertaken

1. Qualified NPTEL exam for the course “Introduction to Large Language Models (LLMs)” in April, 2025.
2. Online FDP Certification of ATAL Academy, AICTE for the course “Introduction to Generative AI Models and Applications” conducted by R.V.R. & J.C. College of Engineering in January, 2025.
3. Online FDP Certification of ATAL Academy, AICTE for the course “Advanced Recurrent Neural Network (RNN) for Sequential Data” conducted by Park College Of Engineering And Technology in February, 2025.
4. Online FDP Certification of ATAL Academy, AICTE for the course “Artificial Intelligence” conducted by ABES Engineering College in December, 2020.
5. Online certification course of “Comprehensive SAS Training” conducted over Infosys Springboard platform.
6. Attended the workshop on Artificial Intelligence and Deep Learning conducted by LeadingIndia.ai in collaboration with Bennett University
7. Attended the workshop for image recognition and object detection conducted by LeadingIndia.ai in collaboration with NVidia and Bennett University
8. Attended the workshop on Emerging Areas in Differential Equations and Real-World Application jointly organized by Indian Mathematical Society, Gujarat Council on Science and Technology, DST and DAIICT
9. Online Certification of the course “DEEP LEARNING EXPLAINED” in EDx, conducted by Microsoft
10. Online Certification of the course “Neural Networks and Deep Learning” in Coursera, conducted by deeplearning
11. Online Certification for course “Machine Learning Using Python”, conducted by E&ICT Academy, IIT Kanpur.
12. Online Certification for course “Introduction to Internet of Things”, conducted by E&ICT Academy, IIT Kanpur

Declaration

Here by, I declare that all above mentioned information are true to the best of my knowledge.

Yours Sincerely,

(Nishith A Kotak)
(April 01, 2026)